

Capstone Project: Maruti Suzuki India

Sales Dashboard — July 2026

Pipeline: Excel (raw data) → MySQL Workbench (cleaning) → Power BI (dashboard)

1. Project Objective

Build an end-to-end analytics project that tells the story of Maruti Suzuki's sales performance in India in July 2026 — model-wise, segment-wise, region-wise, and channel-wise (domestic / OEM / export) — with month-on-month and year-on-year comparisons.

Real-world context to ground the project (July 2026 reports):

- Total sales: 241,421 units (domestic + OEM + exports), up ~33.7% YoY from 180,526 units in July 2025
- Domestic sales hit an all-time high of 200,123 units, crossing 2 lakh units in a single month for the first time
- Exports: 30,056 units; OEM supply: 11,242 units
- Utility Vehicles segment: 78,851 units (Brezza, Ertiga, Fronx, Grand Vitara, Victoris, e Vitara, Invicto, Jimny, XL6)
- Passenger cars: ~103,456 units
- Top-selling models: Dzire, Wagon R, Ertiga, Swift, Fronx, Baleno — several crossing 20,000+ units
- Mini segment (Alto, S-Presso) had the fastest YoY growth (~85%) though smallest by volume

2. Phase 1 — Dataset (Excel)

Structure the raw Excel file with a clean, analysis-ready layout — one row per transaction/dispatch record. See the accompanying sample file **Maruti_Sales_Raw_Sample.xlsx** for a ready-to-use practice dataset with realistic columns.

Column	Example	Notes
Sale_Date	2026-07-14	daily granularity within July 2026
Model	Dzire, Wagon R, Ertiga, Swift, Fronx, Baleno, Brezza...	
Segment	Mini / Compact / UV / Van	
Fuel_Type	Petrol / CNG / Hybrid	
Transmission	Manual / AMT / Automatic	
State	Maharashtra, UP, Karnataka, etc.	for state-wise map
Dealer_City		optional, finer granularity
Channel	Domestic / OEM / Export	
Units_Sold	integer	
Ex_Showroom_Price	numeric (Rs.)	for revenue estimate
Prev_Year_Units	integer	July 2025 baseline, for YoY calc

Tip: the sample file deliberately includes messy spots (blank cells, inconsistent text case, duplicate rows, mixed date formats) — that's what the MySQL cleaning phase is for.

3. Phase 2 — Data Cleaning in MySQL Workbench

1. Create schema & import

Create a database `maruti_sales_july2026`. Use Table Data Import Wizard to load the Excel (export to

2. Inspect and profile

```
SELECT COUNT(*) FROM raw_sales;
SELECT COUNT(*) - COUNT(DISTINCT CONCAT(Sale_Date,Model,Dealer_City,Units_Sold))
  AS dup_count FROM raw_sales;
SELECT * FROM raw_sales WHERE Units_Sold IS NULL OR Model IS NULL;
```

3. Standardize text fields

```
UPDATE raw_sales SET Fuel_Type = TRIM(UPPER(Fuel_Type));
UPDATE raw_sales SET Model = TRIM(Model);
```

4. Fix data types & dates

```
UPDATE raw_sales SET Sale_Date = STR_TO_DATE(Sale_Date, '%d-%m-%Y')
  WHERE Sale_Date LIKE '__-__-__';
ALTER TABLE raw_sales MODIFY Sale_Date DATE;
```

5. Handle nulls

```
DELETE FROM raw_sales WHERE Units_Sold IS NULL;
UPDATE raw_sales SET Transmission = 'Manual' WHERE Transmission IS NULL;
```

6. Remove duplicates

```
CREATE TABLE clean_sales AS
SELECT DISTINCT * FROM raw_sales;
```

7. Build dimension tables

Create a lookup/dimension table for segments (Mini, Compact, UV, etc.) and a `fact_sales` table ref

8. Validate

Sanity-check totals against the real anchor numbers (e.g. does `SUM(Units_Sold)` land near 241,421 total / ~200,123 domestic).

4. Phase 3 — Power BI Dashboard

Connect: Get Data → MySQL Database → point to maruti_sales_july2026, load fact_sales + dimension tables.

Data model: Star schema — fact_sales in the center, dimension tables for dim_model, dim_segment, dim_state, dim_date.

Key DAX measures:

```
Total Units = SUM(fact_sales[Units_Sold])
YoY Growth % = DIVIDE([Total Units] - SUM(fact_sales[Prev_Year_Units]),
    SUM(fact_sales[Prev_Year_Units]))
Domestic Share % = DIVIDE(CALCULATE([Total Units],
    fact_sales[Channel]="Domestic"), [Total Units])
Avg Price = AVERAGE(fact_sales[Ex_Showroom_Price])
```

Suggested dashboard pages/visuals:

- Overview page — KPI cards (Total Units, Domestic Units, Exports, YoY %), trend line across July dates, donut by Channel
- Model performance — bar/column chart ranking models by units, table with MoM/YoY %
- Segment analysis — stacked bar by Segment (Mini/Compact/UV etc.)
- Geography — India state map (filled/shape map) showing state-wise units
- Filters/slicers — Model, Segment, Fuel Type, Channel, Date range

5. Deliverables Checklist

- Raw Excel dataset (with intentional messiness documented)
- MySQL script file with all cleaning queries, commented
- Cleaned table(s) / star schema in MySQL
- Power BI (.pbix) file connected live to MySQL, or with a documented refresh method
- 1-page written summary of insights

6. Suggested Timeline (7–10 days)

Day	Focus
1	Finalize dataset structure, source/simulate data, export to Excel
2	Set up MySQL schema, import raw data, profile for issues
3-4	Write and run cleaning queries, build clean fact/dimension tables
5	Validate cleaned data against real July 2026 totals
6-7	Build Power BI data model, write DAX measures
8	Design and build dashboard visuals/pages
9	Polish formatting, add slicers/tooltips, test interactivity
10	Write insights summary, prepare presentation/demo

7. How to Test Yourself Along the Way

- After cleaning: query row counts and totals — do they match public July 2026 figures?
- After modeling: check relationships in Power BI's Model view — no ambiguous many-to-many paths?
- After DAX: cross-check one measure manually in Excel/SQL against the Power BI card value
- Before presenting: can you explain why each cleaning step was needed, not just what it did?